MILITARY SPECIFICATION

PROJECTILE, PRACTICE, 155MM, M804
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U. S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing for the Projectile, Practice, 155MM, M804.

2. APPLICABLE DOCUMENTS

2.1 Government Documents

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defence Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

FSC 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (see 6.8)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

8797088 - Spacer
9331794 - Projectile, 155MM: Practice M804, Assembly and Marking Diagram
9331807 - Packaging and Marking for Pallet, Wood, Ammunition for Projectile, 155MM, Practice, M804
9340773 - Liner, Suppl. Chg., Smoke
9354433 - Alternate Packaging and Marking for Pallet, Wood, Ammunition, for Projectile, 155MM, Practice, M804

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documets forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and ther supplement thereto, if applicable.

ANSI B46.1 - Surface Testure

(Application for copies should be addressed to the American National Stantards Institute (ANSI), 1430 Broadway, NY, NY 10018).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and specifications.

3.2 Projectile assembly. The Projectile Assembly and components shall comply with all requirements specified on Drawing (Dwg.) 9331794 and with all requirements specified in applicable specifications.

3.2.1 Smoke canister. There shall be no foreign matter or smoke composition on the exterior of the canister.

3.3 Proving ground. The projectiles, when tested as specified, shall show no loss or break-up of metal components. The fuzed projectile shall on impact, produce a smoke signature visible across 4000 meters of level terrain.

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3.4 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship. The requirements for workmanship are as shown on the applicable drawings and specifications and the following:

3.5.1 Surface preparation. Preparation of metal surfaces shall be as specified on the applicable drawings and specifications. Cleaning and preparation are required by component details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.5.2 Painting. Painting shall comply with the requirements of the applicable drawings and specifications. All paint shall be dry to the touch before packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.5.3 Surface finish. The requirements for surface finish are as detailed on the applicable drawing. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determinations.

3.5.4 Threads. Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing.

3.5.6 Burr. No part shall have a burr which might interfere with the assembly or function of the projectile or which might be injurious to personnel handling the item.

3.5.7 Foreign matter. No part or assembly shall contain dirt, grease, rust, corrosion or other foreign matter.

3.5.8 Protection of rotating band. The machined rotating band shall be protected from being nicked, burred or otherwise damaged in subseqeunt handling prior to shipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Projectile, 155MM, Practice M804 Assembly and Marking Diagram	7931794	25
Liner, Suppl Chg, Smoke	9340773	5
Spacer	8798088	5

All assemblies and components shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized, by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted for each new contract or whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.3.2 Inspections to be performed. The First Article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to any or all the requirements of the applicable drawings and specifications and to the inspections of Table I. All inspections listed in Table I shall be classified Major unless otherwise specified in referenced paragraphs and specifications.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Projectile Assemblies and Components				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile, 155MM, Practice M804, Assembly and Marking Diagram (Dwg. 9331794)				
	Examination for Defects (Except Proving Ground Test)	5		3.2	4.4.2
	Proving Ground Test	20		3.3	4.5.1
	Components				
	Liner, Suppl Chg, Smoke (Dwg. 9340773)				
	Examination for Defects	5		3.2	4.4.2.1

NOTES:

DRSMC-DA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.4 Quality Conformance Inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot shall contain loaded projectile of only one weight zone.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALIITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70447 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Liner, Suppl. Chg, Smoke Prior to Assembly to Projectile		9340773
			NEXT HIGHER ASSEMBLY 9331299

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Runout of ID		0.40%	3.2	Gage
101	ID, max.		0.40%	3.2	Gage
103	Wall thickness, min.		0.40%	3.2	Gage
104	Thickness of base, min.		0.40%	3.2	Gage
105	Thread pitch diameter, min.		0.40%	3.2	Gage
106	Thread major diameter, min.		0.40%	3.2	Gage
107	Length, max.		0.40%	3.2	Gage
108	Length of thread, min.		0.40%	3.2	Gage
Minor					
201	Any feature missing		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Projectile and Liner Assembly Prior to Installation of Smoke Canister				9331794
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Depth of liner, min.		0.40%	3.2	Gage
102	Depth of liner, max.		0.40%	3.2	Gage
103	Stakes missing or improper, three places		0.40%	3.2	Visual
104	Inside diameter of liner after staking, entire length, min.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70447 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8798088
4.4.2.3	Spacer				NEXT HIGHER ASSEMBLY 9331794
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Height		0.40%	3.2	Gage
102	OD min.		0.40%	3.2	Gage
103	ID max.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Projectile, 155MM: Practice M804 Assembly and Marking Diagram	1 OF 1	9331794
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Projectile zone weight incorrect		100%	3.2	Balance
102	Zone stenciling incorrect		100%	3.2	Visual
103	Any weight zone punch mark inadequate		0.40%	3.2	Visual/Manual
104	Diameter of bourrelet, max (applies if bourrelet has been retouched)		100%	3.2	Gage
105	Lifting plug gasket missing		0.40%	3.2	Visual
106	Spacer missing		0.40%	3.2	Visual
107	Torque test failure of lifting plug		0.40%	3.2	Test
Minor					
201	Grommet missing, damaged or improperly assembled		0.65%	3.2	Visual
202	Protective coating with total damaged area in excess of 645 square mm (1 square inch)		0.65%	3.10.2	Visual
203	Marking misleading or unidentifiable (except for zones), including incorrect color code		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70447 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Packaging and Marking		9331807
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pallet cover damaged or improperly assembled		0.40%	3.2	Visual
102	National Stock Number or DOD Code missing or unidentifiable		0.40%	3.2	Visual
103	Projectile of different zones on the same pallet		0.40%	3.2	Visual
104	Strapping or wire missing or loose		0.40%	3.2	Manual
Minor					
201	Number of projectile assemblies incorrect		0.65%	3.2	Visual
202	Marking misleading or unidentifiable		0.65%	3.2	Visual
203	Strapping or wire improperly engaged		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.5 Methods of inspection.

4.5.1 Proving ground. These tests shall be performed at a Government proving ground in accordance with the applicable acceptance test procedure. The sample projectiles selected shall be fired in a M185 or M199 Series cannon using a M119A1, Zone 8 charge. Chamber pressures will be based on copper pressure gages. The projectiles and propelling charges shall be uniformly temperature conditioned to 21° Celsius (70°F) plus or minus 1° Celsius (2°F) for a minimum of 24 hours prior to testing. Pressure-time traces and differential pressures shall be obtained for information only by means of a spindle tap in the breech of the cannon and a piezo gage at the projectile base location for each projectile fired. Adequate photographic and instrumental coverage shall be provided to assure compliance with requirements. Range and deflections shall be recorded for informational purposes only. In the event of any abnormal condition, as observed by personnel or instrumentation, an attempt shall be made to recover and examine the metal parts for informational purposes. An observer shall be stationed 4000 ± 200 meters from the edge of the impact area to record presence of smoke signature. The test shall be conducted on level terrain and the wind velocity shall not exceed 15 knots at the time of testing.

5. PACKAGING

5.1 Packing and marking. The projectile shall be packed in accordance with Dwg. 9331807 or Dwg. 9354433.

5.2 Marking. Marking shall be in accordance with Dwg. 9331794.

6. NOTES

6.1 Intended use. The projectile covered by this specification is intended for use as a target practice round for 155MM Howitzers.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection procedures and equipment designs for approval - See MIL-A-48078. Submit equipment designs as required to Commander, ARDC, ATTN: DRSMC-QAR-I(D), Dover, New Jersey 07801.

6.4 Inspection lot size. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for

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delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.5 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristics may be measured or gaged to determine acceptability.

6.6 Submission of ballistic test data and ammunition data cards. In additon to the normal distribution of records for items procured by the Department of the Army, two copies of all ballistic test data and ammunition data cards shall be forwarded to Commander, ARDC, ATTN: DRSMC-QAR-Q (D), DRSMC-LCU-CP (D), Dover, New Jersey 07801 and DRSMC-QAD-P(R), Army Master Depository, Rock Island, IL 61299..

6.7 Alternative testing procedures. Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible Government technical agency that their effectiveness is equal to, or better than the quality assurance provisions specified herein.

6.8 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as US Army Armament Research & Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC(D).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-70447 (AR)	PROJECTILE, PRACTICE, 155MM, M804 LOADING, ASSEMBLING AND PACKING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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